Curriculum Vitae
Henry Kvinge

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AI researcher currently focused on AI for math with 6 years of experience developing and leading projects aimed at accelerating AI-enabled scientific discovery and advancing scientifically grounded approaches to making AI systems safer, more secure, more reliable, and more transparent.

Research interests: • AI to accelerate scientific and mathematical discovery • AI safety, security, and robustness • Science of deep learning • New mathematical tools and paradigms to understand AI systems

Education and positions

2021 –	**Affiliate Assistant Professor**, Mathematics Department, University of Washington
2019 –	**AI Researcher**, National Security Directorate, Pacific Northwest National Laboratory
2017-2019	**Postdoctoral Fellow**, Pattern Analysis Lab, Mathematics Department, Colorado State University
2011-2017	**Ph.D. in Mathematics,** University of California, Davis
2004-2010	**B.S. in Mathematics, B.A. in Biochemistry,** University of Washington, *Magna Cum Laude*

Select publications

- Chau, H., Jenne, H., Brown, D., He, J., Raugas, M., Billey, S. C., & **Kvinge, H.** *Machine learning meets algebraic combinatorics: A suite of datasets to accelerate AI for mathematics research*. (2025). International Conference on Machine Learning **(spotlight oral)**.
- He, J., Jenne, H., Chau, H., Brown, D., Raugas, M., Billey, S., & **Kvinge, H.** (2024). *Machines and mathematical mutations: Using GNNs to characterize quiver mutation classes*. (2025). International Conference on Machine Learning.
- Jenne, Helen, Herman Chau, Davis Brown, Jackson Warley, Timothy Doster, and **Henry Kvinge**. *Can we count on deep learning: Exploring and characterizing combinatorial structures using machine learning.* In The 3rd Workshop on Mathematical Reasoning and AI at NeurIPS'23. 2023.
- **Kvinge, H.**, Emerson, T., Jorgenson, G., Vasquez, S., Doster, T., & Lew, J. (2022). *In what ways are deep neural networks invariant and how should we measure this?* Advances in Neural Information Processing Systems, 35, 32816-32829.
- Godfrey, C., Brown, D., Emerson, T., & **Kvinge, H.** (2022). *On the symmetries of deep learning models and their internal representations*. Advances in Neural Information Processing Systems, 35, 11893-11905.

Technical leadership: Served as principal investigator on 10 multi-year projects in AI assurance and AI for science ⋄ Average number of direct reports per project: 8

Highlights

- Led 2 year investment in AI for advanced manufacturing with materials scientist Keerti Kappagantula. Resulted in 13 papers, including ICLR and NeurIPS maintrack work. Publicity: *Machine Learning Accelerates Development of Advanced Manufacturing Techniques*
- Led 3 multiyear research efforts in AI safety and security. Produced tools now widely used in model safety.
- Led an interdisciplinary team of mathematicians and AI researchers to produce the first set of open problems aimed evaluating AI system's conjecturing ability from examples alone. Resulted in 4 papers, including an ICML poster and an ICML oral.

Selected workshop and conference organization

- Organizer, *AMS Special Session on Machine Learning to Accelerate Mathematical Discover* at the 2025 Joint Math Meetings

- Organizer, *Geometry-grounded Representation Learning and Generative Modeling Workshop* at ICML 2024
- Organizer, *Topology, Algebra, and Geometry in Machine Learning Workshop* at ICML 2023

Additional publications

Organized by topic area and then chronologically.

AI for Math and Pure Math:

- Vargas, Max, Helen Jenne, Davis Brown, and **Henry Kvinge**. *Graph neural networks for tensor product decompositions of Lie algebra representations.* In 2nd AI for Math Workshop@ ICML 2025.
- **Kvinge, Henry**, Can Ozan Oğuz, and Michael Reeks. *The center of the twisted Heisenberg category, factorial Schur Q-functions, and transition functions on the Schur graph.* Journal of Algebraic Combinatorics 52, no. 4 (2020): 469-504.
- **Kvinge, Henry**, Anthony M. Licata, and Stuart Mitchell. *Khovanov's Heisenberg category, moments in free probability, and shifted symmetric functions.* Algebraic Combinatorics 2, no. 1 (2019): 49-74.
- **Kvinge, Henry**. *Coherent systems of probability measures on graphs for representations of free Frobenius towers.* arXiv:1810.11555 (2018).
- Kessler, Ian Holm, **Henry Kvinge**, and James B. Wilson. *A Frobenius-Schreier-Sims algorithm to tensor decompose algebras.* arXiv:1812.03346 (2018).
- **Kvinge, Henry**, and Monica Vazirani. *A combinatorial categorification of the tensor product of the Kirillov-Reshetikhin crystal $B^{1,1}$ and a fundamental crystal.* Algebras and Representation Theory 21, no. 6 (2018): 1277-1331.

AI model robustness, safety, and security:

- Yeats, Eric, Darryl Hannan, **Henry Kvinge**, Timothy Doster, and Scott Mahan. *Automating evaluation of diffusion model unlearning with (vision-) language model world knowledge.* In ICML 2025 Workshop on Machine Unlearning for Generative AI.
- Mahan, Scott, Davis Brown, Timothy Doster, and **Henry Kvinge**. *What makes a machine learning task a good candidate for an equivariant network?* In ICML 2024 Workshop on Geometry-grounded Representation Learning and Generative Modeling.
- Munikoti, Sai, Ian Stewart, Sameera Horawalavithana, **Henry Kvinge**, Tegan Emerson, Sandra E. Thompson, and Karl Pazdernik. *Generalist multimodal AI: A review of architectures, challenges and opportunities.* arXiv:2406.05496 (2024).
- Brown, Davis, Charles Godfrey, Cody Nizinski, Jonathan Tu, and **Henry Kvinge**. *Robustness of edited neural networks.* In ICLR 2023 Workshop on Mathematical and Empirical Understanding of Foundation Models. 2023.
- Konz, Nicholas, Charles Godfrey, Madelyn Shapiro, Jonathan Tu, **Henry Kvinge**, and Davis Brown. *Attributing Learned Concepts in Neural Networks to Training Data.* NeurIPS Workshop on Attributing Model Behavior at Scale **(Oral)**. arXiv:2310.03149 (2023).
- Godfrey, Charles, **Henry Kvinge**, Elise Bishoff, Myles Mckay, Davis Brown, Tim Doster, and Eleanor Byler. *How many dimensions are required to find an adversarial example?* In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 2353-2360. 2023.
- Ly, Cuong, Grayson Jorgenson, Dan Rosa de Jesus, **Henry Kvinge**, Adam Attarian, and Yijing Watkins. *ColMix–A simple data augmentation framework to improve object detector performance and robustness in aerial images*. arXiv:2305.13509 (2023).
- Brown, D., & **Kvinge, H.** (2023). *Making corgis important for honeycomb classification: Adversarial attacks on concept-based explainability tools*. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (pp. 620-627).
- **Kvinge, Henry**, Brett Jefferson, Cliff Joslyn, and Emilie Purvine. *Sheaves as a framework for understanding and interpreting model fit.* In Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 4222-4230. 2021.
- Mahan, Scott, **Henry Kvinge**, and Tim Doster. *Rotating spiders and reflecting dogs: a class conditional approach to learning data augmentation distributions.* arXiv:2106.04009 (2021).
- Mahan, Scott, Tim Doster, and **Henry Kvinge**. *DNA: Dynamic network augmentation.* arXiv:2112.09277 (2021).

- **Kvinge, Henry**, Scott Howland, Nico Courts, Lauren A. Phillips, John Buckheit, Zachary New, Elliott Skomski et al. *One representation to rule them all: Identifying out-of-support examples in few-shot learning with generic representations.* arXiv:2106.01423 (2021).
- Bynum, Lucius, Timothy Doster, Tegan H. Emerson, and **Henry Kvinge**. *Rotational equivariance for object classification using xView.* In IGARSS 2020-2020 IEEE International Geoscience and Remote Sensing Symposium, pp. 3684-3687. IEEE, 2020.

Analysis of deep learning model internals:

- **Kvinge, Henry**, Elizabeth Coda, Eric Yeats, Davis Brown, John Buckheit, Sarah McGuire Scullen, Brendan Kennedy et al. *Probing the limits of mathematical world models in LLMs.* In ICML 2025 Workshop on Assessing World Models.
- Yeats, Eric, John Buckheit, Sarah McGuire Scullen, Brendan Kennedy, Loc Truong, Davis Brown, Bill Kay, Cliff Joslyn, Tegan Emerson, Michael J. Henry, John Anthony Emanuello, **Henry Kvinge**. *What do geometric hallucination detection metrics actually measure?.* In ICML 2025 Workshop on Reliable and Responsible Foundation Models.
- Scullen, Sarah McGuire, Davis Brown, Robert Jasper, **Henry Kvinge**, and Helen Jenne. *Permutations as a testbed for studying the effect of input representations on learning.* In ICML 2025 Workshop on Methods and Opportunities at Small Scale.
- Brown, Davis, Madelyn Ruth Shapiro, Alyson Bittner, Jackson Warley, and **Henry Kvinge**. *Wild comparisons: A study of how representation similarity changes when input data is drawn from a shifted distribution.* In ICLR 2024 Workshop on Representational Alignment. 2024.
- Brown, Davis, Charles Godfrey, Nicholas Konz, Jonathan Tu, and **Henry Kvinge**. *Understanding the inner-workings of language models through representation dissimilarity.* In Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, pp. 6543-6558. 2023.
- McGuire, Sarah, Shane Jackson, Tegan Emerson, and **Henry Kvinge**. *Do neural networks trained with topological features learn different internal representations?.* In NeurIPS Workshop on Symmetry and Geometry in Neural Representations, pp. 122-136. PMLR, 2023.
- **Kvinge, Henry**, Davis Brown, and Charles Godfrey. *Exploring the representation manifolds of stable diffusion through the lens of intrinsic dimension.* In ICLR 2023 Workshop on Mathematical and Empirical Understanding of Foundation Models.
- Godfrey, Charles, Elise Bishoff, Myles Mckay, Davis Brown, Grayson Jorgenson, **Henry Kvinge**, and Eleanor Byler. *Testing predictions of representation cost theory with CNNs.* arXiv:2210.01257 (2022).

Mathematical tools for AI:

- Godfrey, Charles, Michael G. Rawson, Davis Brown, and **Henry Kvinge**. *Fast computation of permutation equivariant layers with the partition algebra.* In ICLR 2023 Workshop on Physics for Machine Learning.
- **Kvinge, Henry**, Grayson Jorgenson, Davis Brown, Charles Godfrey, and Tegan Emerson. *Internal representations of vision models through the lens of frames on data manifolds.* In NeurIPS Workshop on Symmetry and Geometry in Neural Representations, pp. 75-115. PMLR, 2024.
- Myers, Audun, **Henry Kvinge**, and Tegan Emerson. *TopFusion: Using topological feature space for fusion and imputation in multi-modal data.* In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 600-609. 2023.
- Courts, Nico, and **Henry Kvinge**. *Bundle networks: Fiber bundles, local trivializations, and a generative approach to exploring many-to-one maps.* In International Conference on Learning Representations (2022).
- Coda, Elizabeth, Nico Courts, Colby Wight, Loc Truong, WoongJo Choi, Charles Godfrey, Tegan Emerson, Keerti Kappagantula, and **Henry Kvinge**. *Fiber bundle morphisms as a framework for modeling many-to-many maps.* In Topological, Algebraic and Geometric Learning Workshops 2022, pp. 79-85. PMLR, 2022.
- Jorgenson, Grayson, **Henry Kvinge**, Tegan Emerson, and Colin Olson. *Random filters for enriching the discriminatory power of topological representations.* In Topological, Algebraic and Geometric Learning Workshops 2022, pp. 183-188. PMLR, 2022.

- **Kvinge, Henry**, Zachary New, Nico Courts, Jung H. Lee, Lauren A. Phillips, Courtney D. Corley, Aaron Tuor, Andrew Avila, and Nathan O. Hodas. *Fuzzy simplicial networks: A topology-inspired model to improve task generalization in few-shot learning.* In AAAI Workshop on Meta-Learning and MetaDL Challenge, pp. 77-89. PMLR, 2021.
- **Kvinge, Henry**, Colby Wight, Sarah Akers, Scott Howland, Woongjo Choi, Xiaolong Ma, Luke Gosink, Elizabeth Jurrus, Keerti Kappagantula, and Tegan H. Emerson. *Dataset to dataspace: A topological-framework to improve analysis of machine learning model performance.* Uncertainty & Robustness in Deep Learning, 2021. arXiv:2107.04714
- **Kvinge, Henry**, Elin Farnell, Jingya Li, and Yujia Chen. *Rare geometries: revealing rare categories via dimension-driven statistics.* In 2019 18th IEEE International Conference On Machine Learning And Applications (ICMLA), pp. 276-281. IEEE, 2019.

AI for Materials Science:

- Tipton, Cody, Elizabeth Coda, Davis Brown, Alyson Bittner, Jung Lee, Grayson Jorgenson, Tegan Emerson, and **Henry Kvinge**. *Haldane bundles: A dataset for learning to predict the Chern number of line bundles on the torus.* In NeurIPS Workshop on Symmetry and Geometry in Neural Representations, pp. 55-74. PMLR, 2024.
- Howland, Sylvia, Keerti-Sahithi Kappagantula, **Henry Kvinge**, and Tegan Emerson. *Invertible temper modeling using normalizing flows and the effects of structure preserving loss.* In Geometry-grounded Representation Learning and Generative Modeling Workshop (GRaM) at ICML 2024, pp. 202-211. PMLR, 2024.
- Brown, Davis, Cody Nizinski, Madelyn Shapiro, Corey Fallon, Tianzhixi Yin, **Henry Kvinge**, and Jonathan H. Tu. *Model editing for distribution shifts in uranium oxide morphological analysis.* In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 8310-8315. 2024.
- Howland, Scott, Lara Kassab, Keerti Kappagantula, **Henry Kvinge**, and Tegan Emerson. *Parameters, properties, and process: Conditional neural generation of realistic SEM imagery toward ML-assisted advanced manufacturing.* Integrating Materials and Manufacturing Innovation 12, no. 1 (2023): 1-10.
- Truong, Loc, WoongJo Choi, Colby Wight, Elizabeth Coda, Tegan Emerson, Keerti Kappagantula, and **Henry Kvinge**. *Differential property prediction: A machine learning approach to experimental design in advanced manufacturing.* In TMS Annual Meeting & Exhibition, pp. 587-595. Cham: Springer Nature Switzerland, 2023.
- Kassab, Lara, Scott Howland, **Henry Kvinge**, Keerti Sahithi Kappagantula, and Tegan Emerson. *TopTemp: Parsing precipitate structure from temper topology.* In Topological, Algebraic and Geometric Learning Workshops 2022, pp. 199-205. PMLR, 2022.

Other AI for Science:

- Horawalavithana, Sameera, Sai Munikoti, Ian Stewart, and **Henry Kvinge**. *SCITUNE: Aligning large language models with scientific multimodal instructions.* arXiv:2307.01139 (2023).
- Feng, Song, Emily Heath, Brett Jefferson, Cliff Joslyn, **Henry Kvinge**, Hugh D. Mitchell, Brenda Praggastis et al. *Hypergraph models of biological networks to identify genes critical to pathogenic viral response.* BMC bioinformatics 22, no. 1 (2021): 287.
- Aminian, Manuchehr, Helene Andrews-Polymenis, Jyotsana Gupta, Michael Kirby, **Henry Kvinge**, Xiaofeng Ma, Patrick Rosse, Kristin Scoggin, and David Threadgill. *Mathematical methods for visualization and anomaly detection in telemetry datasets.* Interface Focus 10, no. 1 (2020): 20190086.

Dimensionality reduction:

- Blumstein, Mark, and **Henry Kvinge**. *Multi-dimensional scaling on groups.* In Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 4239-4244. 2021.
- Chepushtanova, Sofya, Elin Farnell, Eric Kehoe, Michael Kirby, and **Henry Kvinge**. *Dimensionality reduction.* In Data Science for Mathematicians, pp. 291-337. Chapman and Hall/CRC, 2020.
- **Kvinge, Henry**, Elin Farnell, Michael Kirby, and Chris Peterson. *A GPU-oriented algorithm design for secant-based dimensionality reduction.* In 2018 17th International Symposium on Parallel and Distributed Computing (ISPDC), pp. 69-76. IEEE, 2018.
- **Kvinge, Henry**, Elin Farnell, Michael Kirby, and Chris Peterson. *Too many secants: a hierarchical approach to secant-based dimensionality reduction on large data sets.* In 2018 IEEE High Performance Extreme Computing Conference (HPEC), pp. 1-7. IEEE, 2018.

- **Kvinge, Henry**, Elin Farnell, Michael Kirby, and Chris Peterson. *Monitoring the shape of weather, soundscapes, and dynamical systems: a new statistic for dimension-driven data analysis on large datasets.* In 2018 IEEE international conference on big data (big data), pp. 1045-1051. IEEE, 2018.

Compressive sensing and hyperspectral data:

- Skomski, Elliott, Aaron Tuor, Andrew Avila, Lauren Phillips, Zachary New, **Henry Kvinge**, Courtney D. Corley, and Nathan Hodas. *Prototypical region proposal networks for few-shot localization and classification.* arXiv:2104.03496 (2021).
- Farnell, Elin, **Henry Kvinge**, John P. Dixon, Julia R. Dupuis, Michael Kirby, Chris Peterson, Elizabeth C. Schundler, and Christian W. Smith. *A data-driven approach to sampling matrix selection for compressive sensing.* In Computational Imaging V, vol. 11396, p. 1139603. SPIE, 2020.
- Farnell, Elin, **Henry Kvinge**, Julia R. Dupuis, Michael Kirby, Chris Peterson, and Elizabeth C. Schundler. *Total variation vs L_1 regularization: a comparison of compressive sensing optimization methods for chemical detection.* In Computational Imaging V, vol. 11396, pp. 64-76. SPIE, 2020.
- **Kvinge, Henry**, Elin Farnell, Julia R. Dupuis, Michael Kirby, Chris Peterson, and Elizabeth C. Schundler. *More chemical detection through less sampling: amplifying chemical signals in hyperspectral data cubes through compressive sensing.* In Algorithms, Technologies, and Applications for Multispectral and Hyperspectral Imagery XXVI, vol. 11392, pp. 175-185. SPIE, 2020.
- Dupuis, Julia R., John P. Dixon, Elizabeth Schundler, S. Chase Buchanan, J. D. Rameau, David Mansur, **Henry Kvinge**, Elin Farnell, Chris Peterson, and Michael Kirby. *LWIR compressive sensing hyperspectral sensor for chemical plume imaging.* In Chemical, Biological, Radiological, Nuclear, and Explosives (CBRNE) Sensing XXI, vol. 11416, pp. 42-55. SPIE, 2020.
- **Kvinge, Henry**, Michael Kirby, Chris Peterson, Chad Eitel, and Tod Clapp. *A walk through spectral bands: using virtual reality to better visualize hyperspectral data.* In International Workshop on Self-Organizing Maps, pp. 160-165. Cham: Springer International Publishing, 2019.
- Farnell, Elin, **Henry Kvinge**, Michael Kirby, and Chris Peterson. *Endmember extraction on the Grassmannian.* In 2018 IEEE Data Science Workshop (DSW), pp. 71-75. IEEE, 2018.

Selected talks and poster presentations

Organized by topic area and then chronologically.

2025 July International Conference on Machine Learning
(Oral) Machine learning meets algebraic combinatorics: A suite of datasets to accelerate AI for mathematics research

2025 July International Conference on Machine Learning
(Poster) Machines and mathematical mutations: Using GNNs to characterize quiver mutation classes

2025 April Washington State University Seminar: Math and Stats of AI
(Talk) Two topics in AI for math and math for AI

2025 Jan. Clarifai Research Seminar
(Talk) Symmetries in neural networks and why they matter

2025 Jan. Joint Mathematics Meetings Special Session on Geometric and Combinatorial Methods in Deep Learning Theory
(Talk) Symmetries in neural networks and why they matter

2024 Dec. 4th MATH-AI Workshop at NeurIPS'24
(Poster) Machine learning meets algebraic combinatorics: A suite of datasets to accelerate AI for mathematics research

2024 Dec. 4th MATH-AI Workshop at NeurIPS'24
(Poster) Machines and mathematical mutations: Using GNNs to characterize quiver mutation classes

2024 Oct. Harvard University Center for Mathematical Sciences and Applications
(Talk) Developing machine learning datasets for problems in algebraic combinatorics

2024 July Work on Diagrammatic Intuition and Deep Learning in Mathematics
(Talk) Two vignettes in AI for math and math for AI

2023 March Boston College Math and Machine Learning seminar
(Talk) A mathematical lens on the inner workings of deep learning models

2022 Dec. NeurIPS 2022 Workshop on Symmetry and Geometry in Neural Representations
(Talk) Do neural networks trained with topological features learn different internal representations?

2022 Dec. NeurIPS 2022
(Poster) In what ways are deep neural networks invariant and how should we measure this?
2022 Dec. NeurIPS 2022
(Poster) On the symmetries of deep learning models and their internal representations
2022 Aug. DANGER: Data, Numbers, and Geometry
(Talk) How deep learning is being made more robust, more domain-aware, and more capable of generalization through the influence of algebra and topology
2022 April SIAM PNW minisymposium
(Talk) Fiber bundles as a framework for deep learning-based analysis of many-to-one processes
2022 April ICLR 2022
(Poster) Bundle networks: fiber bundles, local trivializations, and a generative approach to exploring many-to-one maps
2022 Oct. Western Washington University and PNNL Seminar on Data-Driven Discovery
(Talk) Mathematical frameworks for comprehensively exploring process parameter/property relationships in advanced manufacturing
2021 July Applied Category Theory 2021
(Talk) Sheaves as a framework for understanding and interpreting model fit
2021 July ICML 2021 Workshop on Uncertainty & Robustness in Deep Learning
(Poster) Dataset to dataspace: A topological-framework to improve analysis of machine learning model performance
2021 Feb. AAAI 2021 Workshop on Meta-Learning
(Poster) Fuzzy simplicial networks: A topology-inspired model to improve task generalization in few-shot learning
2019 Nov. UC Davis Algebra Seminar
(Talk) Using diagrammatics to motivate coherent systems on towers of Frobenius algebras
2019 June Conference on Geometric Data Analysis *(Talk) Representation theory and multidimensional scaling*
2019 June Tensors: Algebra-Computation-Applications,
(Talk) Letting symmetry guide reduction: representation theory and dimensionality reduction
2018 Dec. IEEE International Conference on Big Data,
(Talk) Monitoring the shape of weather, soundscapes, and dynamical systems: a new statistic for dimension-driven data analysis on large data sets
2018 Nov. University of Colorado Lie Theory Seminar,
(Talk) Coherent systems of probability measures on graphs for representations of free Frobenius towers
2018 Oct. Workshop on Representation Theory, Combinatorics, and Geometry
(Talk) Heisenberg categories, towers of algebras, and up/down-transition functions
2018 Sep. Conference: 2018 IEEE High Performance Extreme Computing Conference
(Talk) Too many secants: a hierarchical approach to secant-based dimensionality reduction on large data sets
2018 June Conference: Interactions of quantum affine algebras with cluster algebras, current algebras and categorification
(Talk) Heisenberg categories, towers of algebras, and symmetric functions
2018 May University of Washington Combinatorics Seminar
(Talk) Symmetric functions, towers of algebras, and Heisenberg categories
2018 May University of Colorado Algebraic Lie Theory Seminar
(Talk) Symmetric functions, towers of algebras, and centers of Heisenberg categories
2018 March Pacific Northwest Combinatorics Day
(Flash talk) Centers of Heisenberg categories, symmetric functions, and the combinatorics of induction/restriction functors
2017 Dec. Future Directions in Representation Theory, University of Sydney
(Poster) The center of the twisted Heisenberg category, factorial Schur P-functions, and up/down transition functions on the Schur graph
2017 Oct. University of Colorado Algebraic Lie Theory Seminar
(Talk) The Kirillov-Reshetikhin crystal $B^{1,1}$ and cyclotomic quiver Hecke algebras
2017 Sept. University of Virginia Algebra Seminar
(Talk) Khovanov's Heisenberg category, the asymptotic representation theory of symmetric groups, and shifted symmetric functions
2017 Sept. Rocky Mountain Combinatorics Seminar - Colorado State University
(Talk) Khovanov's Heisenberg category, moments in free probability, and shifted symmetric functions

2017 July Formal Power Series and Algebraic Combinatorics Conference (FPSAC), London
(Talk) Khovanov's Heisenberg category, moments in free probability, and shifted symmetric functions
2016 Oct. AMS Sectional - University of St. Thomas, Minneapolis (invited talk)
Special Session on Combinatorial Representation Theory
(Talk) A surprising connection between Khovanov's Heisenberg category and the asymptotic representation theory of symmetric groups.
2016 Sept. Arizona State University Discrete Math Seminar
(Talk) A graphical calculus for the shifted symmetric functions.
2016 July Formal Power Series and Algebraic Combinatorics Conference (FPSAC), UBC
(Poster) Categorifying the tensor product of the KR crystal $B^{1,1}$ and a fundamental crystal
2016 June US-Mexico Conference on Representation Theory, Categorification, and Noncommutative Algebra, USC
(Poster) Khovanov's Heisenberg category and the asymptotic representation theory of symmetric groups
2016 March University of Oregon, Algebra Seminar
(Talk) The influence of the Kirillov-Reshetikhin crystal $B^{1,1}$ on the structure of simple cyclotomic KLR modules.
2016 Feb. University of Washington, Algebra and Algebraic Geometry Seminar
(Talk) The influence of the KR crystal $B^{1,1}$ on the structure of simple cyclotomic KLR modules.
2016 Jan. UC Berkeley (invited talk)
Berkeley/Davis Combinatorics Gathering
(Talk) The influence of the KR crystal $B^{1,1}$ on the structure of simple cyclotomic KLR modules.
2015 Oct. AMS Sectional - Loyola University, Chicago (invited talk)
Special Session on Combinatorial and Geometric Representation Theory
(Talk) The influence of the KR crystal $B^{1,1}$ on the structure of simple cyclotomic KLR modules.
2015 Oct. UC Davis Algebra and Discrete Math Seminar
(Talk) The influence of the KR crystal $B^{1,1}$ on the structure of simple cyclotomic KLR modules.
2013 Sept. Arizona State University Discrete Math Seminar
(Talk) The Okounkov-Vershik approach to the representation theory of the symmetric group

Funding Information

Available on request

Students, Postdocs Mentored

- Saoirse (née Nico) Courts, *Postdoc, Howard Hugh Medical Institute at Fred Hutch*
- Charles Godfrey, *Senior applied scientist, Thomson Reuters*
- Herman Chau, *PhD student, University of Washington*
- Davis Brown, *PhD student, University of Pennsylvania*
- Sarah Scullen, *Postdoc, Pacific Northwest National Laboratory*
- Scott Mahan, *Data scientist, Pacific Northwest National Laboratory*

Workshops and conferences organized

2025 Dec. Topology, Algebra, and Geometry in Data Science Conference
2025 Jan. Joint Math Meetings Special Session on Machine Learning to Accelerate Mathematical Discovery
2025 Jan. Joint Math Meetings Special Session on Topological, Algebraic, and Geometric Methods for Safe, Robust, and Explainable Machine Learning
2024 July ELLIS Workshop on Geometry-grounded Representation Learning and Generative Modeling at ICML
2023 July 2nd Annual Workshop on Topology, Algebra, and Geometry in Machine Learning at ICML
2023 June Topology, Algebra, and Geometry in Pattern Recognition with Applications at CVPR
2023 Jan. Joint Math Meetings Special Session on Topology, Algebra, and Geometry in the Mathematics of Data Science
2022 July Workshop on Topology, Algebra, and Geometry in Machine Learning at ICML
2022 Jan. Joint Math Meetings Special Session on Geometry in the Mathematics of Data Science
2021 Oct. ICCV Workshop on Topology, Algebra, and Geometry in Computer Vision (Planned)
2021 Jan. Special Session: Geometry in the Mathematics of Data Science, Joint Math Meetings
2021 Jan. Special Session: Mathematics to the Rescue-Addressing Deficiencies in the Analysis of Overhead Imagery Products, Joint Math Meetings

Seminars organized

2021 – 2024: Pacific Northwest Topology, Algebra, and Geometry in Data Science Seminar (Seattle)
2020 – 2022: Western Washington Data-Driven Discovery Seminar (Virtual)
2020 – 2021: Pure Math for Machine Learning (PMML) (Virtual)